$$R- COOH + H_2O \rightleftharpoons RCOO^- + H_3O^+$$

$$Ka = \frac{[RCOO^-][H_3O^+]}{[RCOOH]}$$

The dissociation constant is generally called acidity constant because it measures the relative strength of an acid. The stronger the acid, the higher will be its Ka value.

The dissociation constant of an acid can also be expressed in terms of pKa value.

$$pKa = - \log Ka$$

A stronger acid will have higher Ka value but smaller pKa value.

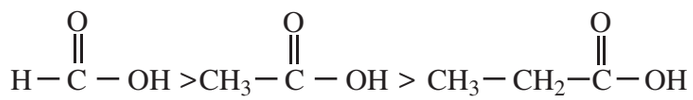
Ka and pKa values of some Carboxylic acids of 298 K

Carboxylic acid		pKa Value
Name of acid	Molecular formula	
Trichloroacetic acid	Cl_3CCOOH	0.64
Dichloroacetic acid	$Cl_2CHCOOH$	1.26
Fluoroacetic acid	FCH_2COOH	2.59
Chloroacetic acid	$ClCH_2COOH$	2.87
Bromoacetic acid	$BrCH_2COOH$	2.90
Iodoacetic acid	ICH_2COOH	3.17
Formic acid	$HCOOH$	3.75
Benzoic acid	C_6H_5COOH	4.20
Acetic acid	CH_3COOH	4.76
Propionic acid	CH_3CH_2COOH	4.88
o – nitrobenzoic acid	$o\text{-}NO_2C_6H_4COOH$	2.17
m-nitrobenzoic acid	$m\text{-}NO_2C_6H_4COOH$	3.49
p- nitrobenzoic acid	$p\text{-}NO_2C_6H_4COOH$	3.44

Effect of substituents on the acidity of carboxylic acid.

i) Electron releasing alkyl group decreases the acidity

The electron releasing groups (+I groups) increase the negative charge on the carboxylate ion and destabilise it and hence the loss of proton becomes difficult. For example, formic acid is more stronger than acetic acid.



Formic acid Acetic acid Propionic acid

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